Algebra 1 Workbook

Polynomials

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ADDING AND SUBTRACTING POLYNOMIALS

1. Which part(s) of the terms stay the same when we add or subtract like terms?

2. Simplify the expression.

$$(2x^3 - 5x^2 + x - 3) - (x^2 - 2x + 7)$$

3. What went wrong in this set of steps?

$$6x^3 + 7 + x^2$$

$$7x^3 + 7$$

4. Simplify the expression.

$$(10a^2b + 3ab^2 - ab) + (2ab^2 - a^2b + ab)$$

5. Simplify the expression.

$$(x^4 - 5y^3 + z - xy) - (2y^4 + 6xy - z + x^4)$$

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6. What went wrong in this set of steps?

$$9 - x^3 + 3 + 4x^3$$

$$12 + 3x^6$$

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MULTIPLYING POLYNOMIALS

1. Use the Distributive Property to expand the expression.

$$\frac{1}{2}(6x+4)(x-1)$$

2. What should we put in place of the "??" to make the expression true?

$$(2x+1)(5-x) = ?? + 10x - x + 5$$

3. What went wrong in this set of steps?

$$(a-2)^2$$

$$a^2 - 4$$

4. Use the Distributive Property to expand the expression.

$$4(2-x)(3+2x)$$

5. Fill in the blank.

$$(3-a)(5+a) = 15 + ___ - a^2$$

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6. Expand the expression.

$(x^2 - 3)(2 - x)$

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DIVIDING POLYNOMIALS

1. Simplify the expression using polynomial long division.

$$(3x^3 - x^2 + 5) \div (x + 2)$$

2. What went wrong in setting up the long division problem?

$$(5x^4 - 3x^2 + x - 2) \div (x^2 + 1)$$

$5x^4 - 3x^2 + x - 2 \;\big|\; x^2 + 1$

3. Express the full solution of the polynomial long division.

$$\begin{array}{r} 3x - 1 \\ x^2 - 3 \;\big|\; 3x^3 - x^2 + x - 5 \\ -(3x^3 + 0x^2 - 9x) \\ \hline -x^2 + 10x - 5 \\ -(-x^2 + 0x + 3) \\ \hline 10x - 8 \end{array}$$

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4. Simplify the expression using polynomial long division.

$$(2x^5 - 3x^3 + x^2 + 4x - 1) \div (x^2 + 2)$$

5. Simplify the expression using polynomial long division.

$$\frac{x^5 - x^3 + 4x^2 - x + 6}{2x^3 - 5}$$

6. Simplify the expression using polynomial long division.

$$(3x^2 + 2x + 5) \div (3x + 5)$$

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MULTIPLYING MULTIVARIABLE POLYNOMIALS

1. Simplify the expression.

$$(a - 3y)(2a + y)$$

2. Simplify the expression.

$$(x - 2y)(x + y) + (3x - y)(4x + 4y)$$

3. Fill in the blanks with the correct terms.

$$(5a - b)(7b - 3a)$$

$$35ab - 15a^2 + ___ + 3ab$$

$$___ - 15a^2 + ___$$

4. What went wrong in this set of steps?

$$(a^2 + 6b)(-a - b^2)$$

$$-a^3 - a^2b^2 - 6ab - b^3$$

$$-a^3 - 7ab - b^3$$

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5. Fill in the the multiplication chart with the correct terms, given the following product of binomials.

$$(4a+3b)(-a+2b^2)$$

		3b
-a		-3ab

6. Simplify the expression.

$$(5ax-3by)(a+y)-(a-y)(2ax+4by)$$

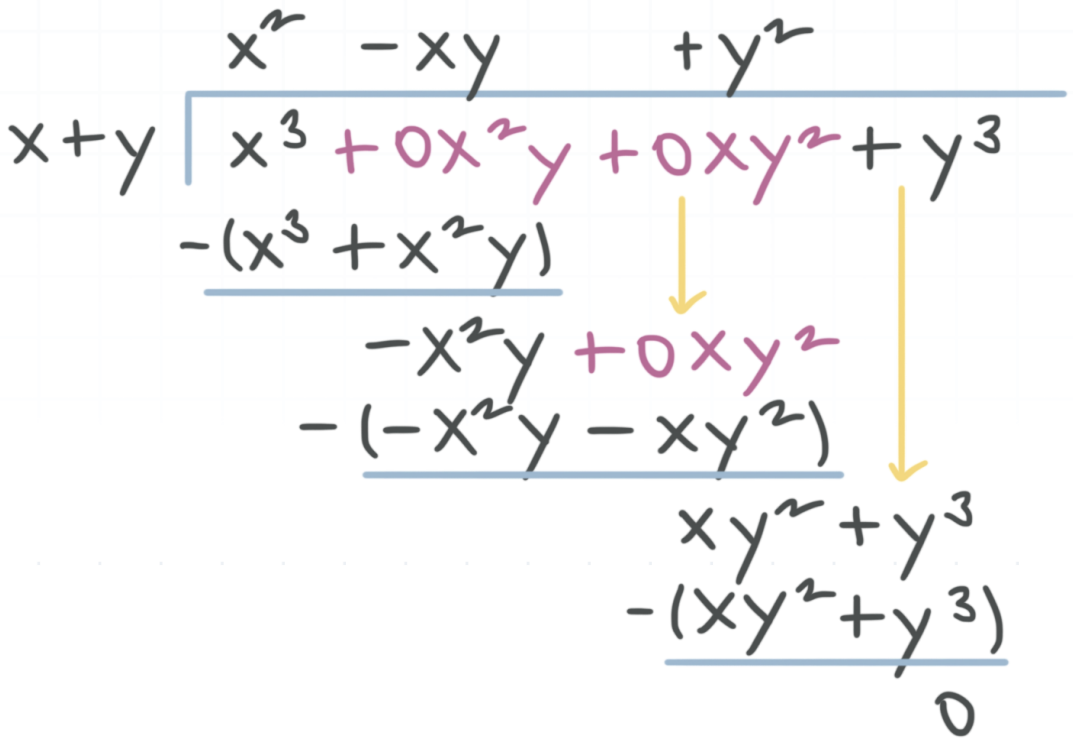
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DIVIDING MULTIVARIABLE POLYNOMIALS

1. Find the quotient.

$$\frac{3x^2 + 6xy - 2y^2}{x - 2y}$$

2. Identify the quotient, remainder, and divisor.



3. How should we rewrite the expression before starting the long division?

$$\frac{2y^3 - xy^2 + x^3}{x - y}$$

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4. Find the quotient.

$$\frac{6x^2 - xy + 2y^2}{2x - y}$$

5. In words, what's the first question we should ask when solving this long division problem?

$2x + 3y \overline{)\ 6x^4 - x^2y + xy^2 + 4y^4}$

6. Find the quotient.

$$(-3x^2 + xy + y^2) \div (x + y)$$

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